

# IRIDA REST API Reference

irida.vigasp.vetinst.no:8080/irida-23.01.3/api

## Authentication

|      |                  |                                                                                                                                                                                                            |
|------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POST | /api/oauth/token | Get bearer token. Params: client_id, client_secret, grant_type=password, username, password. Token valid for configured duration (default 12h). Include in all requests as: Authorization: Bearer {token}. |
|------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Projects

|        |                                         |                                                                                                                                                |
|--------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| GET    | /api/projects                           | List all projects the authenticated user has access to                                                                                         |
| GET    | /api/projects/{id}                      | Get a single project by ID                                                                                                                     |
| GET    | /api/projects/{id}/samples              | List all samples in a project                                                                                                                  |
| GET    | /api/projects/{id}/samples/metadata     | Get metadata for all samples in a project                                                                                                      |
| GET    | /api/projects/{id}/samples/bySampleName | Undocumented endpoint. Requires --location for 302 redirect. Exact name match only, partial names return 404.                                  |
| GET    | /api/projects/{id}/users                | List all users with access to a project                                                                                                        |
| GET    | /api/projects/{id}/analyses             | List analyses shared with a project                                                                                                            |
| GET    | /api/projects/{id}/hash                 | Get project hash - integer checksum for change detection. Store previous hash to detect changes; value alone is not useful without a baseline. |
| POST   | /api/projects                           | Create a new project. Required: name (min 5 chars). Optional: projectDescription, organism                                                     |
| POST   | /api/projects/{id}/samples              | Create a new sample in a project. Required: sampleName (min 3 chars). Optional: organism, description, collectionDate, collectedBy             |
| POST   | /api/projects/{id}/users                | Add a user to a project. Required: userId (username). Optional: role (PROJECT_USER   PROJECT_OWNER)                                            |
| PATCH  | /api/projects/{id}                      | Update project properties. Fields: name, projectDescription, organism                                                                          |
| DELETE | /api/projects/{id}/users/{username}     | Remove a user from a project                                                                                                                   |
| DELETE | /api/projects/{id}/samples/{sampleId}   | Remove a sample from a project. Sample and sequence files are NOT deleted from IRIDA storage.                                                  |

## Samples

|       |                                          |                                                                                                                                                                                    |
|-------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET   | /api/samples/{id}                        | Get full metadata for a single sample                                                                                                                                              |
| GET   | /api/samples/{id}/sequenceFiles          | List all sequence files for a sample (single-end + paired-end). Does not surface pair identifier - call /pairs separately to get pair_id for downloads.                            |
| GET   | /api/samples/{id}/sequenceFiles/pairs    | List paired-end sequence files only                                                                                                                                                |
| GET   | /api/samples/{id}/sequenceFiles/unpaired | ■ BUG (IRIDA 23.01.3): returns HTTP 500. Servlet parses "unpaired" as Long field (NumberFormatException). Workaround: filter /sequenceFiles excluding IDs in /pairs. See issue #4. |
| GET   | /api/samples/{id}/sequenceFiles/fast5    | List FAST5 files for a sample                                                                                                                                                      |
| GET   | /api/samples/{id}/metadata               | Get unstructured key-value metadata for a sample (separate from structured fields at /api/samples/{id}). See issue #7.                                                             |
| GET   | /api/samples/{id}/assemblies             | List assemblies for a sample (populated only by IRIDA pipeline runs)                                                                                                               |
| PATCH | /api/samples/{id}                        | Update sample metadata: sampleName, description, organism, collectionDate, collectedBy, strain, isolate, latitude, longitude, geographicLocationName, isolationSource              |

## Sequence Files

|        |                                                 |                                                                                                                                                                                                                               |
|--------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET    | /api/samples/{id}/pairs/{pairId}/files/{fileId} | Get or download an individual sequence file. Requires Accept: application/fastq to download. fileName carries R1/R2 convention. NOTE: ONLY endpoint where Accept header has effect - all others return text/plain regardless. |
| POST   | /api/samples/{id}/sequenceFiles                 | Upload a single-end FASTQ file. Multipart form field: file=@/path/to/file.fastq.gz                                                                                                                                            |
| POST   | /api/samples/{id}/pairs                         | Upload a paired-end FASTQ pair. Multipart form fields: file1=@R1.fastq.gz file2=@R2.fastq.gz, parameters1=JSON, parameters2=JSON ({miseqRunId, layoutType}). Confirmed working. See bashrc.d issue #27.                       |
| POST   | /api/samples/{id}/sequenceFiles/pairs           | ■ BUG (IRIDA 23.01.3): documented upload endpoint does NOT work for POST - use /api/samples/{id}/pairs instead. Confirmed empirically. See bashrc.d issue #27.                                                                |
| POST   | /api/samples/{id}/sequenceFiles/fast5           | Upload a FAST5 file. Multipart form field: file=@/path/to/file.fast5                                                                                                                                                          |
| DELETE | /api/samples/{id}/sequenceFiles/{fileId}        | Delete a sequence file from a sample                                                                                                                                                                                          |

## Sequencing Runs (admin only)

|        |                         |                                                                                                                                                                        |
|--------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET    | /api/sequencingrun      | List all sequencing runs. Returns empty resources array for non-admin tokens.                                                                                          |
| GET    | /api/sequencingrun/{id} | ■ BUG (IRIDA 23.01.3): HATEOAS link points to /api/sequencingrun/miseqrun which throws HTTP 500 - same routing bug as /unpaired (NumberFormatException). See issue #6. |
| POST   | /api/sequencingrun      | Create a sequencing run. Required: layoutType (SINGLE_END   PAIRED_END), sequencerType (miseq   nextseq   miniseq   directory). uploadStatus auto-set to UPLOADING.    |
| PATCH  | /api/sequencingrun/{id} | Update sequencing run uploadStatus. Values: UPLOADING   COMPLETE   ERROR                                                                                               |
| DELETE | /api/sequencingrun/{id} | Delete a sequencing run. Requires admin token. Confirmed working on VIGASP 23.01.3.                                                                                    |

## Analysis Submissions

|     |                                                     |                                                                                                                                                                                             |
|-----|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET | /api/analysisSubmissions                            | List all analysis submissions for the current user                                                                                                                                          |
| GET | /api/analysisSubmissions/analysisType/phylogenomics | List phylogenomics analyses. ■ SLOW: queries Galaxy synchronously per submission - can saturate all uwsgi workers making IRIDA unresponsive. See issue #8.                                  |
| GET | /api/analysisSubmissions/analysisType/assembly      | List assembly and annotation analyses. ■ SLOW: same Galaxy-blocking behaviour. Expect 15-60+ seconds.                                                                                       |
| GET | /api/analysisSubmissions/analysisType/sistr         | List SISTR typing analyses. ■ SLOW: same Galaxy-blocking behaviour.                                                                                                                         |
| GET | /api/analysisSubmissions/{id}                       | Get a single analysis submission. NOTE: analysisType field is NOT present here - only available via /{id}/analysis when state=COMPLETED.                                                    |
| GET | /api/analysisSubmissions/{id}/analysis              | Get completed analysis outputs (only when state=COMPLETED). Contains analysisType and HATEOAS links to output files. IRIDA ignores Accept header - all output files returned as text/plain. |

## AJAX Endpoints (undocumented, session auth required - no Bearer token)

|      |                                                      |                                                                                                                                                                                                                                                                                      |
|------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AJAX | /ajax/analyses/queue                                 | Get running and queued analysis counts. Returns {"running": N, "queued": N}. Pure DB query via analysisSubmissionRepository.countByAnalysisState() - no Galaxy roundtrip. Auth: @PreAuthorize("hasRole('ROLE_USER')") - any authenticated user. Safe to poll. Discovered 2026-05-06. |
| AJAX | /ajax/analyses/states                                | List all possible AnalysisState enum values with localized labels. Reference data for UI dropdowns.                                                                                                                                                                                  |
| AJAX | /ajax/analyses/types                                 | List registered workflow types (e.g. assembly, phylogenomics, sistr).                                                                                                                                                                                                                |
| AJAX | /ajax/analyses/list                                  | Paginated analysis table. POST with filter/sort/page params. Powers the web UI analyses table.                                                                                                                                                                                       |
| AJAX | /ajax/analyses/{submissionId}/updated-table-progress | Get updated state and duration for a single analysis submission.                                                                                                                                                                                                                     |
| AJAX | /ajax/projects/associated/list?projectId={id}        | List associated projects for a given project. Discovered 2026-04-29 via access log.                                                                                                                                                                                                  |

## Users (documented as admin-only - confirmed accessible with unprivileged token, see issue #5)

|     |                                |                                      |
|-----|--------------------------------|--------------------------------------|
| GET | /api/users                     | List all users in the system         |
| GET | /api/users/{username}          | Get a single user by username        |
| GET | /api/users/{username}/projects | List projects the user has access to |

## Miscellaneous

|     |              |                                                                    |
|-----|--------------|--------------------------------------------------------------------|
| GET | /api         | Root endpoint - returns links to all top-level resources (HATEOAS) |
| GET | /api/version | Get IRIDA server version string. Requires Bearer token.            |